Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CSDE713]

# SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

## DATA WAREHOUSING AND MINING

**Time: 3 Hours** **Maximum Marks:70**

**Instructions:**
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

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| **1.** | **a.** | Differentiate star schema and snowflake schema with examples.. | **07** |
| | **b.** | Differentiate between OLAP and OLTP | **07** |
| **2.** | **a.** | Explain in detail the different steps in knowledge discovery in databases. | **07** |
| | **b.** | Explain the different ways of handling missing values in the data cleaning process of data pre-processing | **07** |
| **3.** | **a.** | Illustrate the working of decision tree classifier with an example. | **08** |
| | **b.** | Explain the advantages and disadvantages of Decision Tree. | **06** |
| | | **OR** | |
| **4.** | **a.** | Demonstrate the working of Naïve Bayes method with an example | **08** |
| | **b.** | Write short notes on:<br>**i.** Confusion matrix<br>**ii.** precision<br>**iii.** recall | **06** |
| **5.** | **a.** | What is clustering? Explain its desirable features. | **07** |
| | **b.** | Demonstrate the working of K-means algorithm with an example. | **07** |
| | | **OR** | |
| **6.** | **a.** | Explain the different applications of clustering | **06** |
| | **b.** | Explain density-based spatial clustering algorithm (DBSCAN) with an example | **08** |
| **7.** | **a.** | What is web mining? Explain its different classification. | **07** |
| | **b.** | Explain the working of page rank algorithm | **07** |
| | | **OR** | |
| **8.** | **a.** | Explain Regression and Trend Analysis in Time-Series Data | **07** |
| | **b.** | Explain how data mining can be used for recommendation system | **07** |